Locally Dense Diophantine Sets in the Rational Subfield

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Abstract

We give an explicit polynomial of degree 13, height 162, and four total coordinates whose existential projection over $\mathbb{Q}(i)$ is a proper subset of $\mathbb{Q}$. This subset is dense simultaneously in every finite collection of completions of $\mathbb{Q}$, and the polynomial is locally soluble for every input in every completion of $\mathbb{Q}(i)$. Nevertheless, the rational input -1 has no global witness. The construction uses the curve $y^2 = x^3 + 9$ and a rank-zero elliptic obstruction. We prove a general simultaneous density statement for ratios of elliptic coordinates under rank stability, and give a bounded-denominator criterion that would suffice for existential definability of a ring of integers. A theorem of Corvaja–Zannier shows that finite rational combinations of points on fixed abelian varieties always miss infinitely many rational integers. We apply this obstruction to the ratio constructions, including a four-coordinate variant of degree 11. The bounded-denominator criterion therefore requires a construction beyond such finite rational combinations.

1 An explicit polynomial and its scope

A subset S of a field L is *Diophantine in* L if membership in S can be expressed by the existence of finitely many solutions of polynomial equations over L. The total number of coordinates includes the distinguished input. The height of an integer polynomial is the largest absolute value of its coefficients.

Put $K = \mathbb{Q}(i)$ and define

$$F(W, U) = U^2W - 9W^3 - 1, \tag{1}$$

$$G(T, W, V) = V^2TW - 9T^3W^3 - 1, \tag{2}$$

$$P(T, W, U, V) = T(F(W, U)^2 - 2G(T, W, V)^2). \tag{3}$$

Let

$$D = \{t \in K : \exists w, u, v \in K,\ P(t, w, u, v) = 0\}. \tag{4}$$

Theorem 1.1. *The following statements hold.*

1. *D is a proper subset of $\mathbb{Q}$, with $0, 1 \in D$ and $-1 \notin D$.*
2. *For every finite nonempty set S of places of $\mathbb{Q}$, the diagonal image of D is dense in $\prod_{v \in S} \mathbb{Q}_v$.*
3. *For every place $\mathfrak{p}$ of K and every $t \in K_\mathfrak{p}$, the equation $P(t, W, U, V) = 0$ has a solution over $K_\mathfrak{p}$.*
4. *The polynomial P has four total coordinates, degree 13, height 162, and 11 nonzero monomials.*

The arithmetic needed for this theorem consists of two small rank-zero calculations, given in Section 2 with reproducible unconditional descent computations. The other parts of the proof are given in Sections 3–5.

Theorem 1.1 concerns a proper subset of the rational subfield. Its coordinate count is therefore not a coordinate bound for a Diophantine definition of $\mathbb{Z}$, $\mathbb{Q}$, or $\mathbb{Z}[i]$ in $\mathbb{Q}(i)$. In particular, local solubility for every input and simultaneous local density do not establish the completeness of an existential definition of the rational subfield.

2 The elliptic curves and the arithmetic input

Consider the elliptic curves

$$E : y^2 = x^3 + 9, \qquad E' : y^2 = x^3 + 81, \qquad E^- : y^2 = x^3 - 9.$$

Their origins are denoted by O.

Lemma 2.1. *One has*

$$E'(\mathbb{Q}) = \{O, (0,9), (0,-9)\}, \qquad E^-(\mathbb{Q}) = \{O\}.$$

The point $R = (3,6) \in E(\mathbb{Q})$ has infinite order.

Proof. The rank-zero statements for E' and E^- are the rigorously computed entries 972b2 and 3888k1 in Cremona's tables; the corresponding entry for E is 972c1 [1, 2, 3]. They can be reproduced independently by unconditional 2-descent. In PARI/GP 2.17.2, the commands

```
ellrank(ellinit([0,0,0,0,81]))
ellrank(ellinit([0,0,0,0,-9]))
```

both return `[0, 0, 0, []]`. The first two entries are proven lower and upper bounds on the Mordell–Weil rank. PARI's bounds use the 2-Selmer group and the Cassels pairing; no Birch–Swinnerton-Dyer assumption or numerical analytic-rank calculation is used [4, `ellrank`]. The accompanying program `check_elliptic_ratio_obstruction.py` repeats these computations.

For completeness, the torsion orders can also be checked by elementary finite-field calculations. Both curves have good reduction at 5 and 7. Direct counts give

	$\#E_*(\mathbb{F}_5)$	$\#E_*(\mathbb{F}_7)$
E'	6	9
E^-	6	7.

Reduction is injective on torsion of order prime to the residue characteristic. These two counts bound the torsion order of E' by 3 and force the torsion subgroup of E^- to be trivial. The three displayed points of E' are distinct and rational, proving the asserted groups once the ranks are zero.

Finally,

$$[2](3,6) = \left(-\frac{15}{16}, \frac{183}{64}\right).$$

The Nagell–Lutz theorem says that a rational torsion point on an integral short Weierstrass equation has integral coordinates [5, Chapter VIII, Section 7]. Thus $(3,6)$ cannot have finite order. □

Lemma 2.2. *Every point of $E(\mathbb{Q}(i))$ is rational:*

$$E(\mathbb{Q}(i)) = E(\mathbb{Q}).$$

Proof. Let σ be complex conjugation and take $A \in E(\mathbb{Q}(i))$. Put $B = A - \sigma A$. Then $\sigma B = -B$. If $B \neq O$, write $B = (a, ib)$ with $a, b \in \mathbb{Q}$: the abscissa is fixed by σ and the ordinate is negated. Its equation gives

$$-b^2 = a^3 + 9, \qquad b^2 = (-a)^3 - 9.$$

Consequently $(-a, b)$ is an affine rational point of E^-, contrary to Lemma 2.1. Therefore $B = O$, so $A = \sigma A$ and $A \in E(\mathbb{Q})$. □

This proof establishes equality of point sets, which is stronger than equality of ranks. There is no unexamined finite-index quotient in the definition used below.

Proposition 2.3. *Let $d \neq 1$ be a squarefree integer. If*

$$E^{(d)} : Y^2 = X^3 + 9d^3$$

has trivial rational Mordell–Weil group, then $E(\mathbb{Q}(\sqrt{d})) = E(\mathbb{Q})$. The hypothesis holds for

$$d \in \{-15, -13, -11, -10, -1, 3, 5, 7\}.$$

Proof. With σ the nonidentity automorphism, an affine point $A - \sigma A$ has the form $(a, b\sqrt{d})$, with $a, b \in \mathbb{Q}$. Its equation $db^2 = a^3 + 9$ gives the rational point $(da, d^2 b)$ on $E^{(d)}$. Triviality of $E^{(d)}(\mathbb{Q})$ thus forces $A = \sigma A$. For each of the eight displayed values of d, the accompanying checker computes matching unconditional rank bounds zero and torsion order one for $E^{(d)}$. □

3 The global projection and the excluded value

Define the set of nonzero abscissas

$$X = \{x(A) : A \in E(\mathbb{Q}),\ A \neq O,\ x(A) \neq 0\}.$$

Proposition 3.1. *The set in (4) is exactly*

$$D = \{0\} \cup X/X.$$

Proof. The element 2 is not a square in $\mathbb{Q}(i)$: if $(a + bi)^2 = 2$ with $a, b \in \mathbb{Q}$, then $ab = 0$, and neither $a^2 = 2$ nor $-b^2 = 2$ is possible. Hence for $t \neq 0$, (3) vanishes if and only if $F = G = 0$.

The equation $F = 0$ forces $w \neq 0$ and is equivalent to the assertion that

$$A = \left(\frac{1}{w}, \frac{u}{w}\right)$$

lies on E. Likewise $G = 0$ forces $tw \neq 0$ and is equivalent to

$$B = \left(\frac{1}{tw}, \frac{v}{tw}\right) \in E.$$

By Lemma 2.2, both points are rational. Their abscissas are nonzero and $x(A)/x(B) = t$, so $t \in X/X$.

Conversely, if $t = x(A)/x(B)$ for points with nonzero abscissas, put $w = 1/x(A)$, $u = y(A)/x(A)$, and $v = y(B)/x(B)$. These rational coordinates satisfy $F = G = 0$. The factor T in (3) supplies the input $t = 0$. □

The chart used in the proof has a simple geometric interpretation. In the projective equation $Y^2 Z = X^3 + 9Z^3$, take the affine chart $X \neq 0$ and put $u = Y/X$, $w = Z/X$. The equation becomes $F = 0$. Thus nonvanishing of the original abscissa is enforced by the affine chart itself, without a separate inverse coordinate. The equation $G = 0$ results from eliminating the second chart coordinate by $w_2 = tw$.

Corollary 3.2. *The same polynomial P defines the same set D in each field $\mathbb{Q}(\sqrt{d})$ listed in Proposition 2.3.*

Proof. In each listed field, 2 is nonsquare: for a squarefree $d \neq 1$, the only quadratic field containing $\sqrt{2}$ is the one with $d = 2$, which is absent from the list. The proof of Proposition 3.1 therefore applies using Proposition 2.3 in place of Lemma 2.2. □

Proposition 3.3. *One has $-1 \notin D$.*

Proof. Otherwise there would be rational points with abscissas a and $-a$, where $a \neq 0$. If their ordinates are b and c, then

$$b^2 = 9 + a^3, \qquad c^2 = 9 - a^3.$$

Multiplying gives

$$(bc)^2 = 81 - a^6 = (-a^2)^3 + 81.$$

Hence $(-a^2, bc) \in E'(\mathbb{Q})$. Lemma 2.1 forces its abscissa to be zero, a contradiction. □

For the size assertion, direct expansion of (3) gives

$$\begin{aligned} P = {} & TU^4W^2 - 18TU^2W^4 - 2TU^2W + 81TW^6 + 18TW^3 - T \\ & - 2T^3V^4W^2 + 36T^5V^2W^4 + 4T^2V^2W - 162T^7W^6 - 36T^4W^3. \end{aligned}$$

The eleven monomials are distinct. Their maximum total degree is 13, attained by T^7W^6, and the maximum coefficient magnitude is 162. The supplied sparse expansion checker verifies these claims with integer arithmetic.

4 Solubility over every completion

Proposition 4.1. *Let L be $\mathbb{R}$, $\mathbb{C}$, or a finite extension of $\mathbb{Q}_p$. For every $t \in L$ the equation $P(t, W, U, V) = 0$ has an L-solution.*

Proof. The assertion is immediate for $t = 0$. Suppose $t \neq 0$. Squares in L contain a neighborhood of the nonzero square 9. For the nonarchimedean fields this follows from Hensel's lemma, and for the archimedean fields it follows from the local square-root function at 9.

Choose $s \in L^\times$ sufficiently close to zero that both $9 + s^3$ and $9 + t^3s^3$ are squares. This gives points of $E(L)$ with nonzero abscissas s and ts, whose ratio is t. The chart substitution in Proposition 3.1 therefore gives a solution with $F = G = 0$. □

In particular, the excluded global input -1 has witnesses over every completion. There are uniform elementary choices for its local abscissa s: use $s = p$ for odd $p \neq 3$, $s = 9$ for $p = 3$, $s = 2$ for $p = 2$, and $s = 1$ over $\mathbb{R}$. For $p \neq 2, 3$ the two radicands reduce to 9 modulo p; for $p = 3$ their quotients by 9 are 1 modulo 3; for $p = 2$ the radicands are 17 and 1. Thus no local obstruction detects this global exclusion.

5 Simultaneous density of global ratios

Local solubility alone does not prove density of the global values. The following lemma supplies the required global approximation.

Lemma 5.1. *Let $E/\mathbb{Q}$ be an elliptic curve, let $R \in E(\mathbb{Q})$ have infinite order, and let S be a finite set of places of $\mathbb{Q}$. There is a positive integer N such that the closure of $\{[Nn]R : n \in \mathbb{Z}\}$ in $\prod_{v \in S} E(\mathbb{Q}_v)$ contains a product of open neighborhoods of O.*

Proof. For each finite $p \in S$, choose an open subgroup of $E(\mathbb{Q}_p)$ on which the elliptic logarithm is an analytic group isomorphism onto $p^{a_p}\mathbb{Z}_p$. After choosing N sufficiently divisible, $[N]R$ belongs to all these subgroups. Its logarithm ℓ_p is nonzero, because the logarithm is injective there and R is not torsion. The closure of its integer multiples is therefore the open subgroup corresponding to $\ell_p\mathbb{Z}_p$.

The Chinese remainder theorem shows that the diagonal integers are dense in $\prod_{p\in S,\ p<\infty}\mathbb{Z}_p$. Thus the integer multiplier can approximate independently any prescribed element of each $\ell_p\mathbb{Z}_p$.

If the real place belongs to S, enlarge N so that $[N]R$ lies in $E(\mathbb{R})^0$. Identify this connected real Lie group with $\mathbb{R}/\mathbb{Z}$. The image α of $[N]R$ is irrational, since otherwise the point would be torsion. Any finite list of p-adic approximation requirements on n defines an arithmetic progression $n = n_0 + Mk$. Along this progression, $n\alpha$ is dense modulo one because $M\alpha$ is irrational. The real approximation can therefore be imposed simultaneously with the finite-place conditions. This proves the assertion. □

Proposition 5.2. *Let $E/\mathbb{Q}$ have positive rank, and let $C \in E(\mathbb{Q})$ be an affine point that is not a point of order two. Put $c = x(C)$ and*

$$X_c = \{x(A) - c : A \in E(\mathbb{Q}),\ A \neq O,\ x(A) \neq c\}.$$

Then X_c/X_c is dense in $\prod_{v\in S}\mathbb{Q}_v$ for every finite nonempty set S of places of $\mathbb{Q}$.

Proof. Fix a point $R \in E(\mathbb{Q})$ of infinite order and apply Lemma 5.1. The rational points $C + [Nn]R$ are dense in a product of neighborhoods of C. The function $f = x - c$ has nonzero derivative at C, since $dx = 2y\,\omega$ for the invariant differential ω, and $y(C) \neq 0$. The local inverse function theorem therefore shows that the values $f(C + [Nn]R)$ are dense in a product $\prod_{v\in S} U_v$, where each U_v is a neighborhood of zero in $\mathbb{Q}_v$.

Take target values $a_v \in \mathbb{Q}_v$. For each v, choose $b_v \in U_v \setminus \{0\}$ sufficiently small that $a_v b_v \in U_v$. Choose independently two global points in the above sequence, with their f-values sufficiently close to $(a_v b_v)_v$ and $(b_v)_v$. Both values can be required to be nonzero: the zeros of f are only C and $-C$, so at most finitely many integer multipliers are excluded. Taking their quotient approximates $(a_v)_v$ as closely as desired. □

Apply Proposition 5.2 to the curve E above and $C = (0, 3)$. Then $c = 0$ and $X_c = X$. Together with Propositions 3.1, 3.3, and 4.1, this completes the proof of Theorem 1.1.

Remark 5.3. The density is in completions of $\mathbb{Q}$. The rational subset D is not asserted to be dense in a completion $K_{\mathfrak{p}}$ in which the closure of $\mathbb{Q}$ is a proper local subfield. Proposition 4.1 is a different assertion: it allows the witnesses themselves to range over the completion.

6 The general rank-stability construction

Ratios of local parameters on a stable abelian variety are used for weak approximation in [11, Theorem 4.2(b)]. In the elliptic case the following statement includes the real place and records an explicit coordinate bound.

Proposition 6.1. *Let $K/\mathbb{Q}$ be a number field and let $E : y^2 = x^3 + ax + b$ be an elliptic curve over $\mathbb{Q}$ with*

$$0 < \operatorname{rank} E(\mathbb{Q}) = \operatorname{rank} E(K).$$

There is a Diophantine subset $D_E \subseteq \mathbb{Q}$ inside K that is simultaneously dense in every finite product of completions of $\mathbb{Q}$. It has a definition with six total coordinates and four equations. If $m \geq 1$ annihilates $E(K)/E(\mathbb{Q})$, the equations have degree at most $\max(3, 3m^2 - 1)$.

Proof. The quotient $E(K)/E(\mathbb{Q})$ is finite by the Mordell–Weil theorem and equality of ranks. Choose m annihilating it and a point $R \in E(\mathbb{Q})$ of infinite order. Put $c = x([m]R)$. Write the abscissa of multiplication by m as

$$x([m]A) = \frac{\phi_m(x(A))}{B_m(x(A))},$$

where $\deg \phi_m = m^2$ and $\deg B_m = m^2 - 1$; these are the usual division-polynomial expressions, with $B_1 = 1$. Put $A_m = \phi_m - cB_m$.

For coordinates $(t, x_1, y_1, x_2, y_2, z)$ impose

$$\begin{aligned} y_j^2 - x_j^3 - ax_j - b &= 0 \qquad (j = 1, 2), \\ tA_m(x_2)B_m(x_1) - A_m(x_1)B_m(x_2) &= 0, \\ zB_m(x_1)B_m(x_2)A_m(x_2) - 1 &= 0. \end{aligned}$$

The last equation excludes every denominator and the zero denominator of the ratio. Each multiplied point is rational, so the resulting input t is rational. The degree bounds are 3, $2m^2$, and $3m^2 - 1$, respectively.

To prove simultaneous density, use rational inputs near R in the sequences provided by Lemma 5.1. The function $x \circ [m] - c$ has a simple zero at R, because the isogeny $[m]$ is unramified in characteristic zero and $[m]R$ is not two-torsion. The proof of Proposition 5.2 applies unchanged. □

This proposition supplies locally rich rational values. It does not give every rational integer. The following theorem rules out that conclusion for every finite rational combination of such values.

7 Finite rational combinations of abelian points

We use the following specialization of [7, Theorem 1.1]. A rational map is evaluated only at points of its domain of definition.

Theorem 7.1 (Corvaja–Zannier)**.** *Let k be a number field. For finitely many semiabelian varieties B_j/k, rational maps $f_j : B_j \dashrightarrow \mathbb{P}^1_k$, and finitely generated subgroups $\Gamma_j \subset B_j(k)$, the set*

$$\mathbb{Z} \setminus \bigcup_j f_j(\Gamma_j)$$

is infinite.

A semiabelian variety is an extension of an abelian variety by a torus; in particular every abelian variety is semiabelian. The next consequence needs only the abelian case.

Corollary 7.2. *Fix a number field k and finitely many abelian varieties over k. Consider finitely many constructions in which one chooses finitely many points on these varieties and applies fixed rational functions to their coordinates, with every division defined. One may also add finitely many specified values. The union of the resulting value sets misses infinitely many rational integers.*

The same conclusion holds if additional conditions restrict the chosen points while leaving each output a fixed rational function of those points.

Proof. For one construction, form the product B of all the abelian varieties supplying its point coordinates, with repeated factors when several independent points are chosen on the same variety. This product is an abelian variety over k. Composition of the fixed rational functions

gives a rational map $f : B \dashrightarrow \mathbb{P}^1$. The tuples on which the original expressions are defined form a subset of its domain; cancellation in a rational expression can enlarge that domain and cannot remove any such tuple. Thus the constructed values are contained in $f(B(k))$. A construction with no point coordinates contributes only a constant.

By the Mordell–Weil theorem, every $B(k)$ is finitely generated. Apply Theorem 7.1 to the finitely many products and maps. Their union misses infinitely many integers. Removing a finite list of specified values from that infinite complement leaves it infinite. Restricting the allowed point tuples only decreases the value sets, so it preserves the conclusion. □

Corollary 7.3. *The sets D of Theorem 1.1 and D_E of Proposition 6.1 each miss infinitely many rational integers. No finite union of fixed sums, products, or ratios of finitely many elements of these sets, even with finitely many values added, can contain $\mathbb{Z}$.*

Proof. For D, Proposition 3.1 gives $D = \{0\} \cup X/X$, where X is the set of nonzero abscissas of $E(\mathbb{Q})$. Every nonzero value is therefore the value of the rational function $x(P_1)/x(P_2)$ on $E \times E$. The defining chart forces both abscissas and the denominator to be nonzero. The stratum $T = 0$ contributes exactly the one additional input value 0.

For D_E, the guard in Proposition 6.1 excludes all poles and the zero denominator of

$$\frac{x([m]P_1) - c}{x([m]P_2) - c},$$

a rational function on $E_K \times E_K$. Hence its entire value set is contained in the rational-map image required by Corollary 7.2.

For a finite rational combination involving D, separate the finitely many possibilities in which one of its arguments is the added value 0. Each remaining argument is supplied by a pair of elliptic points. After substituting the chosen zero values, every nonempty defined case is again a fixed rational expression on a product of elliptic curves, or a constant. There are finitely many such cases, so the preceding corollary applies. □

The same obstruction applies to a smaller-degree slope construction. For any of the eight quadratic fields in Proposition 2.3, put

$$\begin{aligned} P_{\mathrm{sl}}(T, Z, W, S) = T((T^2Z^2W - 9W^3 - 1)^2 \\ - 2(Z^2S - 9S^3 - 1)^2). \end{aligned}$$

Proposition 7.4. *The existential projection of P_{sl} over each of these fields is*

$$D_{\mathrm{sl}} = \{0\} \cup \left\{ \frac{y(P_1)/x(P_1)}{y(P_2)/x(P_2)} : P_1, P_2 \in E(\mathbb{Q}),\ x(P_1)x(P_2) \neq 0 \right\}.$$

It misses infinitely many rational integers. The polynomial has four total coordinates, degree 11*, height* 162*, and eleven nonzero monomials.*

Proof. For a nonzero input t, nonsquareness of 2 forces both squared residuals to vanish. The constant term -1 makes w, s nonzero. The resulting elliptic points are

$$(1/w, tz/w), \qquad (1/s, z/s).$$

Exact point stability makes both points rational. The second ordinate is nonzero, since an ordinate-zero rational point on E would give a rational solution of $x^3 = -9$, which does not exist. Thus $z \neq 0$, and the ratio of the two slopes is t. Conversely any two points in the displayed set give witnesses by taking $w = 1/x(P_1)$, $s = 1/x(P_2)$, and $z = y(P_2)/x(P_2)$. The input 0 is represented separately by the leading factor T, so no extra nonzero stratum arises.

The slope ratio is a rational function on $E \times E$, defined on every pair just used. Corollary 7.2 therefore proves the infinite omission after adding 0. Expansion of the displayed polynomial gives the stated numerical data; it is independently recorded by `check_rationality_filters.py`. □

These results concern outputs given by fixed rational functions of abelian points. They do not cover arbitrary algebraic correspondences whose output is merely algebraic over those point coordinates, or constructions with unrestricted additional affine witnesses. Such a construction requires its own soundness and completeness arguments. The obstruction is consequently a restriction on a class of methods, not a proof that a ring of integers is not Diophantine in its field of fractions.

8 Fibers and rational parametrizations

Proposition 8.1. *For $t \in \mathbb{Q} \setminus \{0, 1\}$, the smooth projective curve associated to*

$$y^2 = s^3 + 9, \qquad z^2 = t^3 s^3 + 9$$

has genus four. In particular, the nonzero-ratio witnesses for each such fixed rational input form a finite set.

Proof. Over an algebraic closure, the function field is a biquadratic extension of $\overline{\mathbb{Q}}(s)$. Each square root has three finite branch points and is ramified at infinity. The two finite branch sets are disjoint because $t^3 \neq 1$. Thus there are seven branch points in their composite, each with ramification index two. Riemann–Hurwitz gives

$$2g - 2 = 4(-2) + 7 \cdot 2 = 6,$$

so $g = 4$. Faltings's theorem gives finiteness of its rational points [6]. The affine nonzero-ratio witnesses are a subset. □

The points with $s = 0$ supply four rational points on the displayed projective curve. They are precisely the points removed when forming a nonzero abscissa ratio. For $t = -1$, the obstruction in Proposition 3.3 excludes every remaining affine point with $s \neq 0$.

Proposition 8.2. *Let k be a characteristic-zero field, let E/k be an elliptic curve, and let $h \in k(E^r)$ for some $r \geq 1$. The rational map $h : E^r \dashrightarrow \mathbb{P}^1$ has no rational section. In particular, there are no $w, u, v \in k(T)$ satisfying $F = G = 0$ as identities for the polynomials in (1)–(2).*

Proof. Every point of $E(k(T))$ is constant: a rational map $\mathbb{P}^1_k \dashrightarrow E$ extends to a morphism by properness, and a nonconstant morphism is impossible by Riemann–Hurwitz. Indeed, after extending scalars to an algebraic closure its ramification formula would read $-2 \geq 0$.

A putative rational section of a ratio therefore has constant elliptic points and hence constant ratio. For the explicit system, $F = G = 0$ would give the two points

$$(1/w, u/w), \qquad (1/(Tw), v/(Tw))$$

over $k(T)$, both constant, while their abscissa ratio is the nonconstant parameter T. This is impossible. □

This obstruction concerns rational parametrizations. It does not exclude an arithmetic procedure that chooses unrelated rational points separately for different inputs.

9 A sufficient bounded-denominator condition for integrality

The following criterion gives a different target for the number-field problem. Unlike local density, it directly suffices for an existential definition of the full ring of integers.

Theorem 9.1 (Integral-basis criterion; cf. [11, Theorem 4.2(d)])**.** *Let $K/\mathbb{Q}$ have degree n, and suppose that $S \subset K$ is Diophantine in K with*

$$\mathbb{Z} \subseteq S \subseteq \mathcal{O}_K.$$

Then $\mathcal{O}_K$ is Diophantine in K. If S has a definition with N total coordinates, one polynomial, and degree d, then $\mathcal{O}_K$ has a definition with nN total coordinates and n equations of degree at most d.

Proof. Choose an integral basis $\omega_1, \dots, \omega_n$ of $\mathcal{O}_K$. The two inclusions in the hypothesis give

$$\mathcal{O}_K = \omega_1 S + \cdots + \omega_n S.$$

Indeed, the right side is integral, and the integer coefficients of every integral-basis expansion belong to S.

For an input x, retain $s_1, \dots, s_{n-1}$ and eliminate $s_n = (x - \sum_{j<n} \omega_j s_j)/\omega_n$. Impose n copies of the definition of S, with disjoint witness blocks. The count is $1 + (n-1) + n(N-1) = nN$; the substitutions are linear and do not increase degree. □

Corollary 9.2. *The conclusion still holds if $\mathbb{Z} \subseteq S$ and $MS \subseteq \mathcal{O}_K$ for some positive rational integer M. With the same notation, nN total coordinates and n equations of degree at most Md suffice.*

Proof. The finite union

$$S' = \bigcup_{j=0}^{M-1} (j + MS)$$

lies in $\mathcal{O}_K$ and contains $\mathbb{Z}$, by division with remainder. If $f(T, \mathbf{Y}) = 0$ defines S, the polynomial

$$\prod_{j=0}^{M-1} M^d f\left(\frac{T-j}{M}, \mathbf{Y}\right)$$

defines S' using the same witness block. A product over a field vanishes if and only if one factor vanishes, so shared witnesses create no additional condition. Its degree is at most Md. Apply Theorem 9.1. □

Neither criterion requires that S be a ring. Neither follows from the simultaneous density proved above. In fact, the set D in Theorem 1.1 has unbounded denominators at every prime, as its density in each $\mathbb{Q}_p$ immediately shows.

10 Counterexample searches and the distinction between ambient rings

The explicit exclusion in this paper is not a counterexample to the existence of an existential definition of $\mathbb{Z}[i]$ in $\mathbb{Q}(i)$. It excludes one proposed way of obtaining the complete rational subfield from stable elliptic coordinates.

The available undefinability results for large, or ample, fields do not provide such a number-field counterexample. A field is large if every smooth curve with a rational point has infinitely many rational points. Fehm's theorem and Kwon's extension show that infinite Diophantine sets in a perfect large field cannot be confined to a proper subfield or even to a finite union of affine translates of proper subfields [8, 9]. No number field is large: a smooth genus-two curve such as $y^2 = x^5 - x + 1$, with point $(0, 1)$, has only finitely many points over each number field

by Faltings's theorem. The polynomial is squarefree, since a common root of $x^5 - x + 1$ and $5x^4 - 1$ would have to be $5/4$, which does not satisfy the derivative equation.

Similarly, the rank-stability transfer theorem of Mazur–Rubin–Shlapentokh defines one ring of integers inside another ring of integers; its witnesses are integral [10, Theorem 1.1 and Section 14]. Its approximate-logarithm method does not by itself remove that witness-domain restriction. For an approach through elliptic ratios, the remaining requirement is a construction that escapes the rational-map restriction of Section 7 and proves either complete coverage of $\mathbb{Q}$ or the bounded-denominator criterion above. Finite rational operations on the stable elliptic values cannot supply the required inclusion of every integer.

11 Reproducible computation

The program `check_elliptic_ratio_obstruction.py` requires `cypari2` and uses PARI's unconditional rank bounds. It also checks the torsion orders, counts the reductions at 5 and 7 by two independent finite-field methods, verifies the displayed doubling formula, expands (3) with exact integer coefficients, and records the local square conditions. The result is stored in `elliptic-ratio-certificate.json`. The commands

```
python3 -m pip install cypari2
python3 check_elliptic_ratio_obstruction.py \
  --output elliptic-ratio-certificate.json
```

reproduce the calculation. The rank-zero bounds, rather than a finite search for points, are the decisive computational inputs. The standard-library program `check_rationality_filters.py` independently expands P_{sl} and records a bounded elliptic-point search. That finite search is not used to prove the infinite omission of integers; the latter follows from Theorem 7.1 and the proved rational-map containment.

References

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